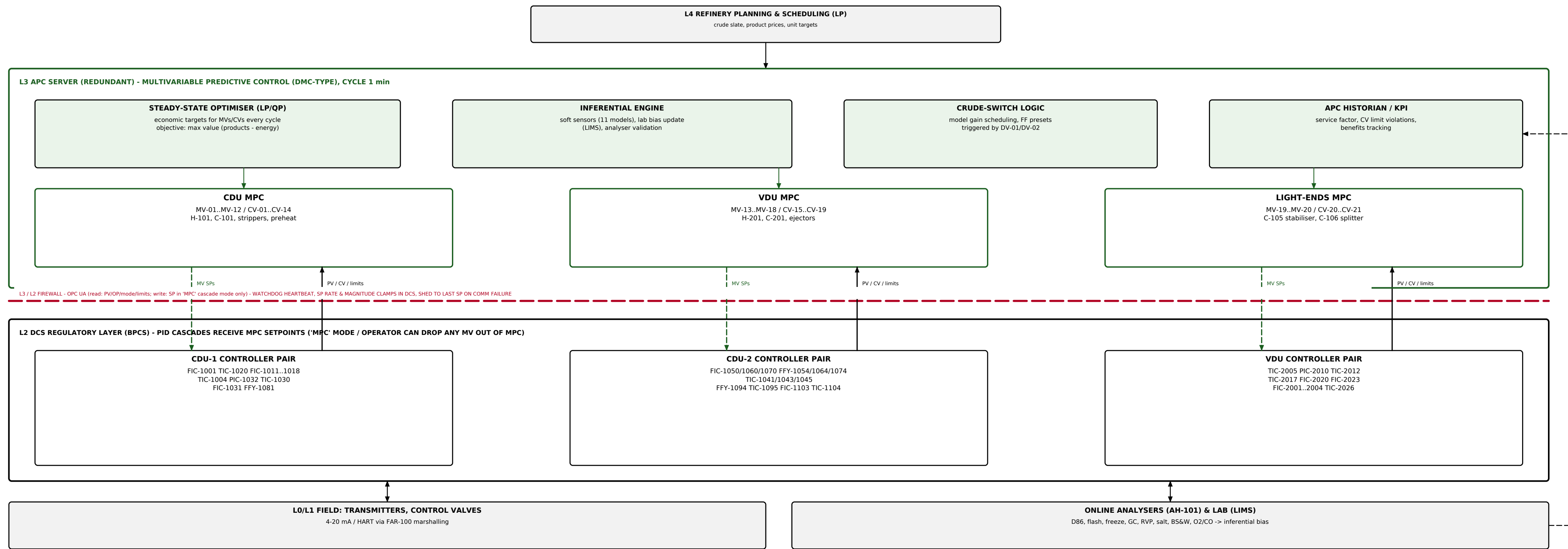




APC DESIGN BASIS

- Three DMC-type MPC applications on a redundant LC3 Aircar (CDU, DUV, lighting ends) with a common steady state LP optimizer; 1 minute execution, prediction horizon = 2 – a longest settling time (CDU = 180 min).
- Three DMC-type MPC applications on a redundant LC3 Aircar (CDU, DUV, lighting ends) after regulatory lighting audit; crude-type gain scheduling (light / heavy loads); 2 minutes execution, prediction horizon = 2 (2 weeks for CDU, 2 weeks for DUV).
- Priority of maximum: (1) safety / regulatory limits (thermal, flooding, wash-off wetting, valve saturation); (2) product specifications; (3) economic optimisation (crank index / HVG0 recovery, minimise energy, wash-out throughput).
- Inferential run in the Aircar server (pressure-compensation temperatures, PCT), validated by analysts and lab with slow bias update; analyser fault – differential with frozen bias; – O2 tripped after 10 min.
- DCS keeps full authority; MPC writes SPs only to temps in "MPC" mode; with SP clamps and rate limits; on watchdog failure all loops shed to AUTO at last good SP and an alarm is raised.
- Expected benefits (industry typical): 1-3 % throughput, 2-5 % energy, 0.5-1 vol % higher-value product recovery.
- MVC/IDV nets are FEED-level; final 1-3 % of the Aircar MPC specification after the pre-test.

MV	DCS TAG (SP)	DESCRIPTION	RANGE	MPC
MV-01	FIC-1001	Crude charge	331-729 m3/h	CDU
MV-02	TIC-1020	H-101 COT	356-372 °C	CDU
MV-03	TIC-1030	C-101 top temperature (naphtha EP)	124-144 °C	CDU
MV-04	FIC-1050	Kerosene draw	±15 % of design	CDU
MV-05	FIC-1060	Diesel draw	±15 % of design	CDU
MV-06	FIC-1070	AGO draw	±20 % of design	CDU
MV-07	TIC-1041	TPA duty (return T)	73-108 °C	CDU
MV-08	TIC-1043	MPA duty (return T)	150-185 °C	CDU
MV-09	TIC-1045	BPA duty (return T)	214-249 °C	CDU
MV-10	FFY-1081	C-101 stripping steam ratio	6-12 lb/bbl AR	CDU
MV-11	FFY-1054/1064/1074	Side-stripper steam ratios	2-8 lb/bbl	CDU
MV-12	PIC-1032	C-101 drum top pressure	0.4-1.0 barg	CDU
MV-13	TIC-2005	H-201 COT	386-402 °C	VDU
MV-14	PIC-2010	C-201 top pressure	15-40 mbar(a)	VDU
MV-15	TIC-2012	C-201 top temperature	55-90 °C	VDU
MV-16	TIC-2017	HVGO PA duty (return T)	±20 °C	VDU
MV-17	FIC-2020	Wash oil (>= min wetting)	min-design x 1.3	VDU
MV-18	FIC-2023	C-201 stripping steam	0.1-5 x design	VDU
MV-19	FFY-1094 / TIC-1095	C-105 L/F, sensitive tray T	±10 % / ±8 °C	LE
MV-20	FIC-1103 / TIC-1104	C-106 reflux, sensitive tray T	±15 % / ±8 °C	LE

CONTROLLED / CONSTRAINT VARIABLES (CV)				
CV	CONTROLLED VARIABLE	MEASUREMENT	TYPE	MPC
CV-01	Naphtha D86 EP	AT-1038 + inferential	range	CDU
CV-02	Kerosene flash point	AT-1055 + inferential	min limit	CDU
CV-03	Kerosene freeze point	inferential (lab bias)	max limit	CDU
CV-04	Diesel D86 T95	AT-1065 + inferential	max / target	CDU
CV-05	Diesel cloud point / flash	inferential (lab bias)	range	CDU
CV-06	AGO D86 T95 / colour	AT-1075 + inferential	max	CDU
CV-07	Kero-diesel / diesel-AGO gap	inferential (5-95 gap)	min	CDU
CV-08	Overflash	FI-1080 / charge	>= 3 vol %	CDU
CV-09	Section dP (flooding)	PD1-1237	max	CDU
CV-10	H-101 firing / tube metal T	FI-1020B OP, TI skin	max	CDU
CV-11	H-101 arch O2 / draft	AIC-1022, PIC-1023	range	CDU
CV-12	Valve positions	FV-1001/1031, TV bypass, PV-1032A	5-90 %	ALL
CV-13	A-101 duty / OH temperature	TI A-101 outlet	max	CDU
CV-14	Desalter inlet T	TIC-1004 OP / PV	125-145 °C	CDU
CV-15	LVGO D86 T95 / HVGO CCR, Ni+V	inferential + lab	max	VDU
CV-16	Wash-bed wetting rate	FIC-2020 / bed area	>= min	VDU
CV-17	C-201 flash zone T / bottoms T	TI flash zone, TIC-2026	max	VDU
CV-18	Ejector load / PV-2010 OP	PIC-2010 OP	max	VDU
CV-19	H-201 firing, coil outlet P	PIC-2006 OP, PI	max	VDU
CV-20	LPG C5+ / naphtha RVP	AT-1099 / TIC-1095 infer.	max	LE
CV-21	LN C6+ / HN C5- (BP)	AT-1108 / AT-1109	max	LE

INFERENTIAL MODELS		
INFERENTIAL (SOFT SENSOR)	MAIN INPUTS	BIAS / VALIDATION
Naphtha D86 EP	TIC-1030, PIC-1032, reflux/draw ratio	AT-1038, lab
Kerosene flash point	Draw T, stripper steam ratio, C-101 P	AT-1055, lab
Kerosene freeze point	Kero draw T, MPA return T, P	lab
Diesel D86 T95	Diesel draw T (PCT), HC partial pressure	AT-1065, lab
Diesel cloud point	D86 T95 infer., crude API	lab
AGO D86 T95	AGO draw T, overflow, steam ratio	AT-1075, lab
Overflow vol %	Wash-zone liquid FI-1080 / charge	-
LVGO D86 T95	C-201 top P, TIC-2012, LVGO PA	lab
HVGO CCR / Ni+V	Wash rate, flash zone T, HVGO cut	lab
Naphtha RVP (C-105)	TIC-1095, P, feed rate	lab
Atm. residue 370 °C-	Flash zone T/P, steam ratio	lab

DISTURBANCE VARIABLES (DV)		
DV	DISTURBANCE	SOURCE
DV-01	Crude switch / tank change	Tank-farm signal (OSBL)
DV-02	Crude API / density, salt, B5&W	AT-1048
DV-03	Crude inlet T to heater (CIT)	TI-1226
DV-04	Ambient air temperature	TI-9005 (air coolers)
DV-05	Fuel-gas Wobbe / LHV	AT-1021 (FG analyser)
DV-06	MPLP steam header pressure	PIC-9004
DV-07	AR rate to VDU	FIC-1083 PV (feed-forward)

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME APC / MPC STRUCTURE (MV / CV / DV)			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	6 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-IC-CSD-006			A